

1. Administrative & Study Identification

Full Study Title: A Phase 3 Study to Investigate the Safety, Tolerability, and Immunogenicity of Revaccinating Pregnant Participants During Subsequent Pregnancies and Persistence of Immunity of a Single Dose of a Bivalent Respiratory Syncytial Virus (RSV) Vaccine **Short Title/Acronym:** Marguerite Study (C3671059) **Protocol Number:** C3671059 **NCT Number:** NCT06866405 **Sponsor Name:** Pfizer **Investigational Product:** RSVpreF (Abrysvo®) / Bivalent RSV Vaccine **Phase of Development:** Phase 3 **Trial Status:** Recruiting **Medical Monitor:** Pfizer Clinical Trial Contact Center

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To investigate the safety, tolerability, and immunogenicity of a second dose of the RSVpreF vaccine when given to pregnant participants during subsequent pregnancies. * **Secondary Objectives:** To evaluate the persistence and duration of immunity derived from a single dose of RSVpreF given during a previous pregnancy by examining non-pregnant participants.

2.2 Background & Rationale Respiratory syncytial virus (RSV) represents a substantial burden of disease and is a serious risk to millions of babies every year, potentially leading to severe respiratory illnesses like pneumonia or bronchiolitis. Pfizer's maternal RSV vaccine (Abrysvo) was approved for use in pregnancy to help protect newborns against RSV until they are 6 months old. However, it is currently unknown how long this maternal protection lasts into later pregnancies or if it is safe and effective to administer the vaccine again (revaccination) in subsequent pregnancies. This study aims to address this knowledge gap.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Placebo-controlled (for specific randomized cohorts) * **Blinding:** Open-label (for known antibody data cohorts) and Double-blind (for the randomized cohort) * **Randomization:** * **Cohort 1:** Pregnant participants with previous RSVpreF from a Pfizer trial and available antibody data will receive a second dose of RSVpreF. * **Cohort 2:** 1:1 Randomization (RSVpreF vs. Placebo) specifically for pregnant participants who received RSVpreF in a previous pregnancy but lack previous antibody data. * **Cohort 3:** Non-

pregnant participants who previously received RSVpreF in a Pfizer trial will not receive any vaccine and will provide blood samples only.

3.2 Planned Enrollment & Duration * **Target** **\$N\$**: 550 participants (including pregnant participants, their infants, and a non-pregnant observational cohort). * **Number of Sites**: Multicenter. * **Estimated Study Start**: [To be determined] * **Estimated Primary Completion**: [To be determined]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Women aged 18 to 49 years. 2. Currently pregnant, between 24 and 36 weeks gestation, and expecting a single infant without known risks for complications (Cohorts 1 & 2). 3. Documented receipt of the RSVpreF or Abrysvo vaccine during a previous pregnancy. 4. Had an ultrasound scan at 18 weeks or later during the current pregnancy with no major fetal problems detected. 5. Agrees to allow the baby to take part in the study and signs the informed consent. 6. **Infant Participants**: Proof that the parent(s) or legal guardian(s) has signed a consent form and agrees to attend scheduled visits and follow the study plan, including laboratory tests and other procedures. 7. **Non-pregnant Participants (Cohort 3)**: Have already received one dose of the RSVpreF vaccine during a previous pregnancy as part of a Pfizer clinical trial, with usable results, and able to sign a consent form.

4.2 Exclusion Criteria 1. Received any approved or experimental RSV vaccine *since* their previous pregnancy. 2. Pre-pregnancy body mass index (BMI) > 40 kg/m². 3. History of a severe adverse reaction to a vaccine or a serious allergic reaction (like anaphylaxis) to any ingredient in the study vaccine. 4. Experiencing current pregnancy issues, or possessing previous pregnancy issues at the time of giving consent. 5. Women who are breastfeeding at the time of enrollment. 6. **Non-pregnant Participants (Cohort 3)**: Received any approved or experimental RSV vaccine after participating in the Pfizer clinical trial, or taking part in other studies with new drugs within 28 days before giving consent or during the study period.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency**: 1 dose of RSVpreF (Abrysvo) or Placebo administered during weeks 32–36 of pregnancy (for applicable cohorts). * **Route of Administration**: Intramuscular (IM) injection. * **Duration of Treatment**: Single dose. Mother and baby are evaluated and monitored until about 6 months after the baby is born.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard prenatal vitamins and routine obstetric care medications. * **Prohibited:** Any other investigational or approved RSV vaccines outside of the study protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Weeks 24 to 36 of pregnancy	Informed Consent, Medical History, Ultrasound Review, Physical Exam
Baseline/Vaccination	Weeks 32 to 36 of pregnancy	Randomization (if applicable), Single Dose Vaccination, Blood Sample Collection, Vital Signs
Interim Follow-up	Post-vaccination to 6 months post-birth	Maternal safety monitoring (4-5 visits total); Infant health and RSV symptom monitoring (up to 4 visits)
End of Study	6 months post-birth	Final physical exams, vital signs, and safety/immunity evaluations for mother and infant

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Definition:** Safety and tolerability of revaccination during subsequent pregnancies. This is measured by: 1. Percentage of pregnant participants reporting local reactions: pain at the injection site, redness, and swelling. 2. Percentage of pregnant participants reporting systemic events: fever, fatigue, headache, vomiting, nausea, diarrhea, muscle pain, and joint pain. 3. Percentage of pregnant participants reporting adverse events. * **Measurement Method:** Participant electronic symptom diaries, routine physical exams, and clinical observation by the study doctor.

7.2 Secondary & Exploratory Endpoints * Proportion of participants achieving neutralizing antibody to RSV A and RSV B at 3 months and 6 months. * Maternal immunogenicity and RSV antibody generation from a second dose. * Persistence of RSV immunity in non-pregnant participants from a single prior dose. * Birth outcomes and transplacental transfer of RSV immunity to the newborn.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Active monitoring via patient diaries (for injection site reactions and systemic events) and routine HCP check-ups for both the mother and infant during the 6-month follow-up period.
- **Serious Adverse Events (SAEs):** Standard reporting timeline for severe pregnancy or fetal complications.

- **Data Monitoring Committee (DMC):** Internal/External independent safety committee monitoring maternal and fetal outcomes throughout the Phase 3 trial.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for overall efficacy; Safety Set for evaluating maternal/infant adverse events; Immunogenicity Set for blood sample analyses.
 - **Sample Size Justification:** Sized adequately for a Phase 3 study (N=550) to establish non-inferior safety and antibody generation profiles compared to the original, single-dose Phase 3 benchmarking.
 - **Handling of Missing Data:** Standard handling for vaccine trials; subjects missing post-vaccination serology may be excluded from per-protocol immunogenicity analyses.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine clinical site monitoring to ensure proper vaccine administration protocol adherence and extensive infant follow-up tracking.
 - **Audit Trail:** Comprehensive eCRF locking, tracking of maternal delivery records, and infant health/hospitalization records.
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11. Study Identifiers & Governance

Primary ID: C3671059 **Registry Number:** NCT06866405 **Sponsor/Lead Organization:** Pfizer **Study Title:** Marguerite / A Phase 3 Study of Revaccination in Subsequent Pregnancies With Bivalent RSV Vaccine and Duration of Protection of a Single Dose **Official Regulatory Title:** A Phase 3 Study to Investigate the Safety, Tolerability, and Immunogenicity of Revaccinating Pregnant Participants During Subsequent Pregnancies and Persistence of Immunity of a Single Dose of a Bivalent Respiratory Syncytial Virus (RSV) Vaccine **Trial Status:** Recruiting

12. Clinical Framework (RSV Vaccine Specific)

Primary Condition: Respiratory Syncytial Virus (RSV) Infection **Diagnosis Threshold:** Preventative Maternal Immunization (N/A) **Medical Methodology:** Active maternal vaccination to induce passive newborn immunity **Optimization Target:** Robust RSVpreF antibody transplacental transfer and sustained maternal immunogenicity

13. Study Procedures & Methodology

Management Pathway: Maternal vaccination pathway (between 32-36 weeks gestation) **Education Protocol (HCP):** Protocol/cohort allocation training and specific safety monitoring for RSV revaccination **Education Protocol (Patient):** Symptom diary training and awareness of RSV symptoms in newborns **Quality Audit Mechanism:** Close tracking of birth outcomes and infant healthcare utilization over the 6-month follow-up

14. Observation & Follow-Up Schedule

Total Duration: ~6 months post-birth **Onsite Visit Cadence:** 4-5 maternal visits; up to 4 infant visits **Remote Monitoring Frequency:** As required for symptom diary check-ins and adverse event reporting **Checklist Review Interval:** Coinciding with scheduled obstetric and pediatric visits

15. Sign-off & Approval

Role	Name	Signature	Date		:--		:--		:--		:--	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]									
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]									
Statistician	[Name]	_____	[DD-MMM-YYYY]									